

NEWTON 5 ENE, MS, USA

WMO: 747710

Lat: 32.3378N

Lon: 89.0703W

Elev: 374

StdP: 14.50

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 63831

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.0	25.3	7.5	8.2	31.2	14.0	11.3	31.3	14.0	46.5	12.6	44.9	1.8	N/A	0.325

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	93.9	75.6	92.1	75.8	90.2	75.5	79.4	88.4	78.4	87.2	77.7	86.4	4.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	143.4	83.4	76.2	138.8	82.4	75.3	134.6	81.5	43.1	88.5	42.1	87.4	41.3	86.4	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.9	9.2	8.2	15.3	96.2	4.3	2.6	12.2	98.1	9.7	99.6	7.3	101.1	4.2	103.0
			13.8	81.8	5.2	1.2	10.1	82.7	7.0	83.3	4.1	84.0	0.3	84.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.5	45.3	49.2	56.3	62.8	70.7	77.5	79.3	79.2	74.7	64.2	53.4	48.1
	DBStd	14.34	10.46	10.00	8.89	7.05	5.79	3.33	2.82	3.43	4.99	8.32	8.78	9.82
	HDD50	621	222	127	41	4	0	0	0	0	0	4	62	161
	HDD65	2511	614	452	290	120	22	0	0	0	3	118	360	532
	CDD50	5538	75	104	237	389	643	825	908	904	741	445	165	102
	CDD65	1953	2	9	21	55	200	375	443	439	294	94	13	8
	CDH74	17511	4	40	176	546	1788	3511	3974	3907	2575	889	84	17
	CDH80	7135	0	1	12	73	590	1572	1800	1792	1046	241	8	0
Wind	WSAvg	3.1	3.8	4.1	3.9	3.5	2.9	2.5	2.3	2.3	2.5	2.8	3.1	3.6
Precipitation	PrecAvg	57.1	6.0	5.5	5.8	5.4	4.3	4.4	5.0	4.6	3.5	3.5	4.1	5.3
	PrecMax	70.2	12.1	12.3	9.2	14.5	9.7	10.1	9.7	9.9	9.6	9.8	10.7	8.4
	PrecMin	36.3	2.8	1.3	0.8	1.3	1.1	0.7	2.4	1.8	0.4	0.0	0.7	1.4
	PrecStd	8.9	2.5	2.8	2.1	3.2	2.1	2.2	1.9	2.1	2.1	2.6	2.5	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.1	79.0	81.8	84.4	90.6	95.6	95.3	97.3	94.2	89.5	81.2	76.7
		MCWB	62.8	66.5	65.3	69.7	72.7	75.2	76.0	77.5	72.6	67.5	67.0	69.1
	2%	DB	70.4	73.9	78.8	82.2	87.9	93.0	93.0	93.9	91.3	85.6	76.3	72.4
		MCWB	63.1	64.7	65.3	68.6	72.5	75.9	76.8	76.3	73.5	69.6	65.4	67.2
	5%	DB	66.9	69.9	75.5	80.0	86.0	90.6	91.0	91.5	89.1	82.7	72.6	68.5
		MCWB	60.6	63.2	63.5	66.7	71.2	75.8	76.7	76.7	73.7	68.0	63.8	64.0
	10%	DB	63.1	66.5	71.8	77.4	83.8	88.5	89.1	89.2	86.5	79.6	69.4	65.3
		MCWB	57.5	60.5	61.6	65.1	70.5	75.3	76.6	76.1	73.4	67.7	61.6	61.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.3	69.0	68.8	73.3	76.0	79.2	80.7	81.2	78.4	76.1	70.8	71.3
		MCDB	70.7	74.3	75.4	80.4	84.0	88.2	88.6	90.5	86.0	82.3	74.7	74.6
	2%	WB	65.0	67.3	67.2	71.3	74.4	78.2	79.3	79.7	77.2	74.3	68.5	68.2
		MCDB	68.6	73.4	74.1	78.0	82.9	87.5	88.5	88.7	85.2	80.9	72.6	72.2
	5%	WB	62.1	65.4	65.5	69.8	73.3	77.4	78.3	78.5	76.1	72.6	66.1	65.8
		MCDB	64.8	70.5	71.9	76.2	82.1	86.8	87.3	87.0	84.3	78.7	69.8	68.4
	10%	WB	58.6	62.6	64.0	68.2	72.3	76.6	77.6	77.6	75.0	70.5	63.7	62.5
		MCDB	62.4	67.0	70.2	74.1	80.6	85.7	86.4	85.7	83.1	76.1	68.1	65.1
Mean Daily Temperature Range	5% DB	MDBR	22.0	22.2	23.6	24.4	22.1	20.8	19.2	19.7	21.9	24.7	24.9	21.2
		MCDBR	23.4	23.9	25.6	25.7	24.6	24.1	21.8	22.8	24.6	26.3	26.3	22.3
		MCWBR	16.2	13.5	12.2	12.1	10.7	8.9	8.4	8.2	9.1	12.0	15.8	15.3
	5% WB	MCDBR	18.2	18.7	19.1	19.8	20.2	20.4	19.1	18.8	19.6	19.4	20.5	18.1
MCWBR		14.2	11.9	10.7	11.2	9.4	8.8	8.2	8.2	8.1	10.0	14.9	14.1	
Clear-Sky Solar Irradiance	taub		0.328	0.329	0.361	0.395	0.427	0.460	0.509	0.508	0.434	0.370	0.344	0.324
	taud		2.507	2.509	2.429	2.331	2.283	2.224	2.117	2.116	2.307	2.455	2.475	2.534
	Ebn at Noon		283	295	291	284	274	264	251	249	265	276	275	279
	Edh at Noon		28	30	35	40	43	45	50	49	39	31	28	25
All-Sky Solar Radiation	RadAvg		869	1033	1395	1751	1928	1991	1899	1775	1587	1327	987	752
	RadStd		77	147	114	140	123	148	102	125	116	175	96	75

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+209	+128

Nomenclature: See separate page